Cài mongo trên 26.04

MongoDB 7.0 repo noble thây cho version 8

Xóa repo version 8 nếu có

```
test3@test3:~$ sudo rm -f /etc/apt/sources.list.d/mongodb-org-8.0.list
```

Add repo version 7

```
test3@test3:~$ curl -fsSL https://www.mongodb.org/static/pgp/server-7.0.asc | \
sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg \
--dearmor
File '/usr/share/keyrings/mongodb-server-7.0.gpg' exists. Overwrite? (y/N) y
test3@test3:~$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 multiverse" | \
sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list
deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 multiverse
```

```
test3@test3:~$ sudo apt update
Hit:1 http://vn.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://vn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://vn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Get:5 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 InRelease [3,005 B]
Get:6 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0/multiverse amd64 Packages [122 kB]
Get:7 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0/multiverse arm64 Packages [120 kB]
Fetched 245 kB in 2s (116 kB/s)
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
test3@test3:~$ apt-cache policy mongodb-org
mongodb-org:
  Installed: (none)
  Candidate: 7.0.37
  Version table:
     7.0.37 500
        500 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0/multiverse amd64 Packages
     7.0.36 500
```

Cài repo version 7

```
test3@test3:~$ sudo apt install -y mongodb-org
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14-gener
  linux-headers-7.0.0-14-generic        linux-modules-7.0.0-14-generic
  linux-image-unsigned-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Installing:
  mongodb-org

Installing dependencies:
  mongodb-database-tools  mongodb-org-database                mongodb-org-mongos  mongodb-org-shell
  mongodb-mongosh         mongodb-org-database-tools-extra  mongodb-org-server  mongodb-org-tools

Summary:
  Upgrading: 0, Installing: 9, Removing: 0, Not Upgrading: 14
  Download size: 189 MB
  Space needed: 636 MB / 12.4 GB available
```

Kiểm tra version và restart service

```
test3@test3:~$ mongod --version
db version v7.0.37
Build Info: {
    "version": "7.0.37",
    "gitVersion": "9d30419d900008ba3ecf2a14546b236f2158b65b",
    "openSSLVersion": "OpenSSL 3.5.5 27 Jan 2026",
    "modules": [],
    "allocator": "tcmalloc",
    "environment": {
        "distmod": "ubuntu2204",
        "distarch": "x86_64",
        "target_arch": "x86_64"
    }
}
```

```
test3@test3:~$ sudo systemctl restart mongod
test3@test3:~$ sudo systemctl status mongod
● mongod.service - MongoDB Database Server
     Loaded: loaded (/usr/lib/systemd/system/mongod.service; disabled; preset: enabled)
     Active: active (running) since Thu 2026-07-02 01:44:00 UTC; 8s ago
 Invocation: bf4995d3ca0c4c328541cc9c7530ffa0
       Docs: https://docs.mongodb.org/manual
   Main PID: 2834 (mongod)
     Memory: 82.5M (peak: 82.5M)
        CPU: 198ms
     CGroup: /system.slice/mongod.service
             └─2834 /usr/bin/mongod --config /etc/mongod.conf
```

Add db từ node chính

```
rs0 [direct: primary] test> use testdb
switched to db testdb
rs0 [direct: primary] testdb> db.testcollection.insertOne({ name: "MongoDB Replica Set", status: "Working" })
{
  acknowledged: true,
  insertedId: ObjectId('6a45c9c452aff977774ebe81')
}
```

Kiểm tra trên node phụ

```
rs0 [direct: secondary] test> use testdb
switched to db testdb
rs0 [direct: secondary] testdb> db.testcollection.find()
[
  {
    _id: ObjectId('6a45c9c452aff977774ebe81'),
    name: 'MongoDB Replica Set',
    status: 'Working'
```

Test failover

Tắt service trên node chính

```
test1@test1:~$ sudo systemctl stop mongod
```

Node 2 làm tự up lên primary

```
    _id: 1,
    name: 'test2:27017',
    health: 1,
    state: 1,
    stateStr: 'PRIMARY',
    uptime: 3155,
    optime: { ts: Timestamp({ t: 1782959119, i: 1 }), t: Long('7') },
    optimeDate: ISODate('2026-07-02T02:25:19.000Z'),
    lastAppliedWallTime: ISODate('2026-07-02T02:25:19.580Z'),
    lastDurableWallTime: ISODate('2026-07-02T02:25:19.580Z'),
    syncSourceHost: '',
    syncSourceId: -1,
    infoMessage: '',
    electionTime: Timestamp({ t: 1782959069, i: 1 }),
    electionDate: ISODate('2026-07-02T02:24:29.000Z'),
    configVersion: 6,
    configTerm: 7,
    self: true,
    lastHeartbeatMessage: ''
  },
```

Test update sau khi tắt service

```
rs0 [direct: primary] testdb> db.testcollection.find()
[
  {
    _id: ObjectId('6a45c9c452aff977774ebe81'),
    name: 'MongoDB Replica Set',
    status: 'Working'
  },
  {
    _id: ObjectId('6a45cc96da239d980f2c83ea'),
    name: 'TEST SERVICE UPDATE',
    status: 'Working'
  }
```

Mở lại node 1

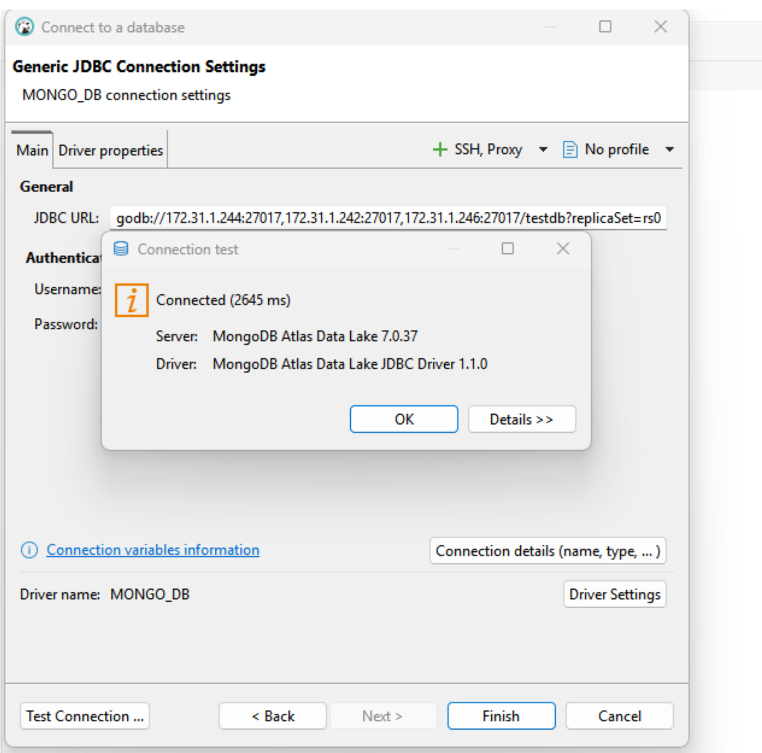
Node 1 lên lại thành node chính đồng thời update từ db

```
rs0 [direct: primary] test> use testdb
switched to db testdb
rs0 [direct: primary] testdb>  db.testcollection.find()
[
  {
    _id: ObjectId('6a45c9c452aff977774ebe81'),
    name: 'MongoDB Replica Set',
    status: 'Working'
  },
  {
    _id: ObjectId('6a45cc96da239d980f2c83ea'),
    name: 'TEST SERVICE UPDATE',
    status: 'Working'
  }
]
```

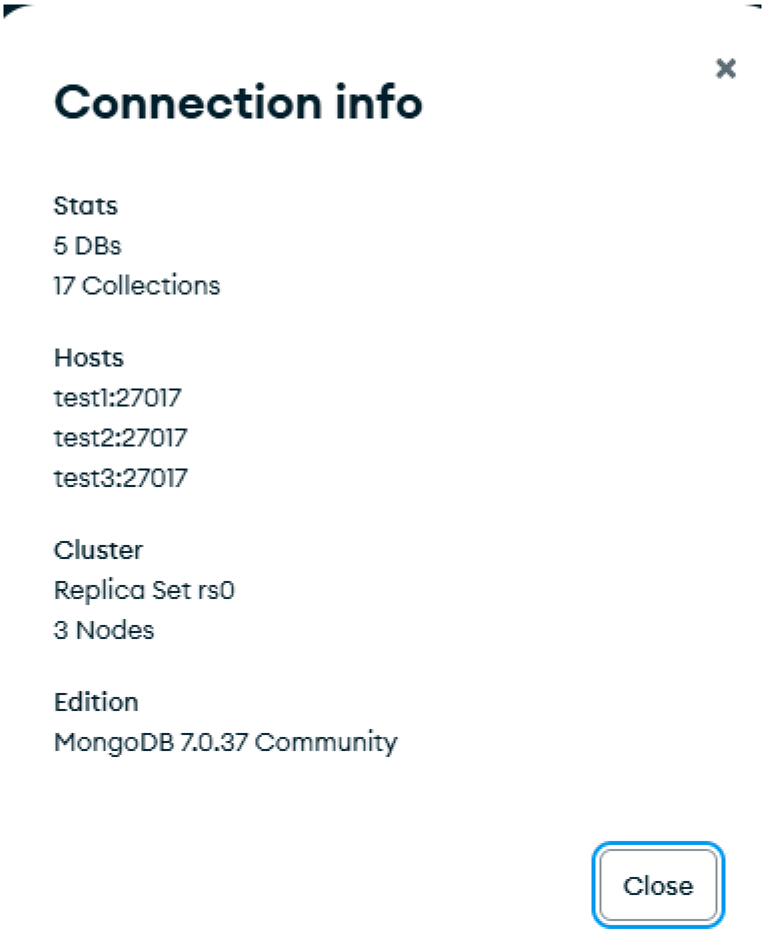
Connect user

```
test2@test2:~$ nc -zv 172.31.1.244 27017
Connection to 172.31.1.244 27017 port [tcp/*] succeeded!
```

```
test3@test3:~$ nc -zv 172.31.1.244 27017
Connection to 172.31.1.244 27017 port [tcp/*] succeeded!
```



DO Driver JDBC của MongoDB **không dùng được với Community Edition**.
GIẢI PHÁP THẤY THẾ DÙNG **MONGODB COMPASS**



```
>_MONGOSH
    {
      _id: 0,
      name: 'test1:27017',
      health: 1,
      state: 1,
      stateStr: 'PRIMARY',
      uptime: 13696,
      optime: { ts: Timestamp({ t: 1782974355, i: 1 }), t: Long('11') },
      optimeDate: 2026-07-02T06:39:15.000Z,
      lastAppliedWallTime: 2026-07-02T06:39:15.625Z,
      lastDurableWallTime: 2026-07-02T06:39:15.625Z,
      syncSourceHost: '',
      syncSourceId: -1,
      infoMessage: '',
      electionTime: Timestamp({ t: 1782960684, i: 1 }),
      electionDate: 2026-07-02T02:51:24.000Z,
      configVersion: 6,
      configTerm: 11,
      self: true,
      lastHeartbeatMessage: ''
    },
    {
      _id: 1,
      name: 'test2:27017',
      health: 1,
      state: 2,
      stateStr: 'SECONDARY',
      uptime: 13684,
      optime: { ts: Timestamp({ t: 1782974355, i: 1 }), t: Long('11') },
```

Khi mới restart Node1 thì Note 1 không ngay lập tức trở thành primary mà vẫn là **secondary**

```
members: [
  {
    _id: 0,
    name: 'test1:27017',
    health: 1,
    state: 2,
    stateStr: 'SECONDARY',
    uptime: 2,
    optime: { ts: Timestamp({ t: 1782975313, i: 1 }), t: Long('12') },
    optimeDurable: { ts: Timestamp({ t: 1782975313, i: 1 }), t: Long('12') },
    optimeDate: 2026-07-02T06:55:13.000Z,
    optimeDurableDate: 2026-07-02T06:55:13.000Z,
    lastAppliedWallTime: 2026-07-02T06:55:13.668Z,
    lastDurableWallTime: 2026-07-02T06:55:13.668Z,
    lastHeartbeat: 2026-07-02T06:55:22.286Z,
    lastHeartbeatRecv: 2026-07-02T06:55:21.307Z,
    pingMs: Long('0'),
    lastHeartbeatMessage: '',
    syncSourceHost: 'test3:27017',
    syncSourceId: 2,
    infoMessage: '',
    configVersion: 6,
    configTerm: 12
  },
  {
    _id: 1,
    name: 'test2:27017',
    health: 1,
    state: 1,
    stateStr: 'PRIMARY',
    uptime: 14649,
    optime: { ts: Timestamp({ t: 1782975313, i: 1 }), t: Long('12') },
    optimeDate: 2026-07-02T06:55:13.000Z,
```

Do đã set up ưu tiên node 1 làm primary nên hệ thống sẽ bầu node 1 thành primary lại!

Nếu không có set up từ trước thì node 2 sẽ vẫn là primary dù node 1 có start lại!

```
},
lastStableRecoveryTimestamp: Timestamp({ t: 1782974693, i: 1 }),
electionCandidateMetrics: {
  lastElectionReason: 'priorityTakeover',
```

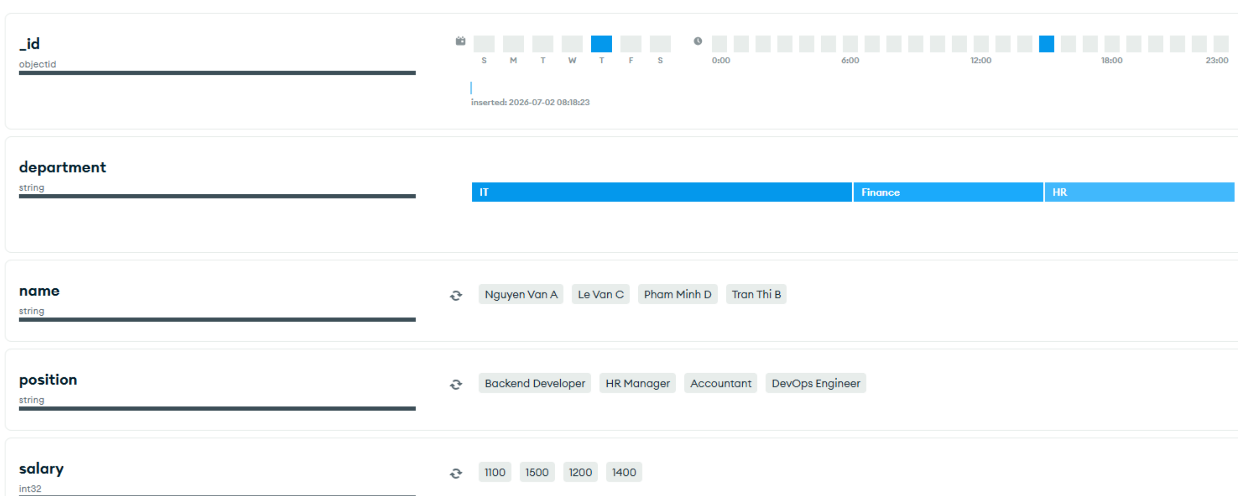
```
members: [
  {
    _id: 0,
    name: 'test1:27017',
    health: 1,
    state: 1,
    stateStr: 'PRIMARY',
```

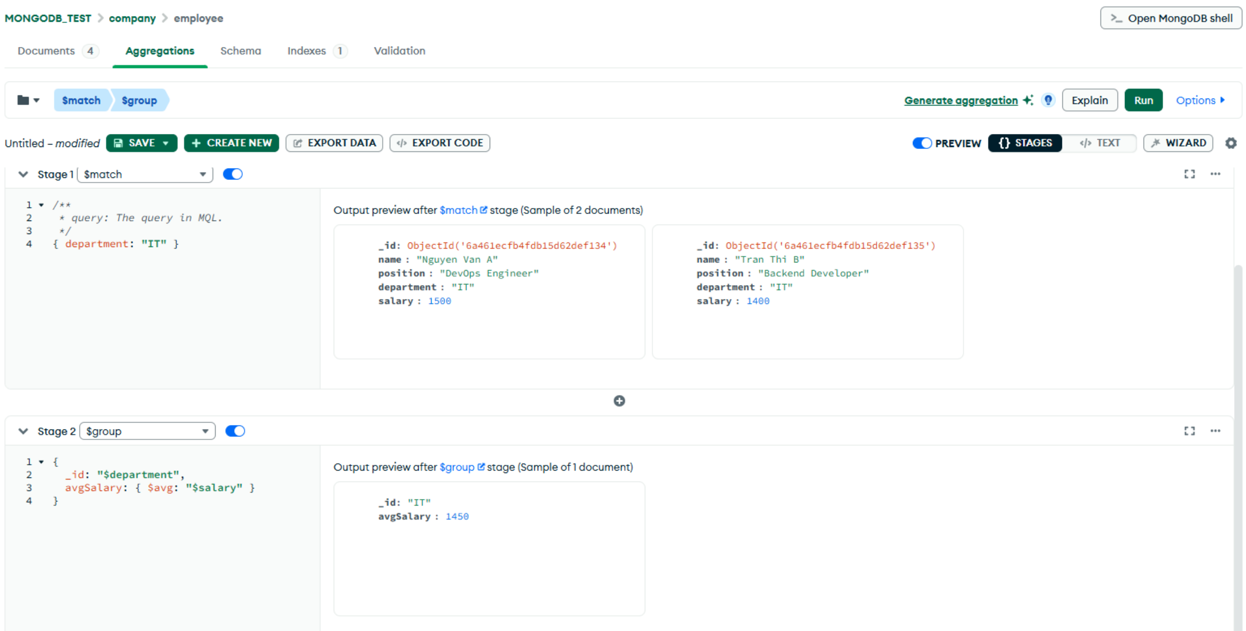
Backup toàn bộ database

```
test1@test1:~$ sudo mongodump --out /backup
2026-07-02T08:57:16.192+0000    writing `admin.system.version` to `/backup/admin/system.version.bson`
2026-07-02T08:57:16.193+0000    done dumping `admin.system.version` (1 document)
2026-07-02T08:57:16.193+0000    writing `test.employee` to `/backup/test/employee.bson`
2026-07-02T08:57:16.193+0000    writing `company.employee` to `/backup/company/employee.bson`
2026-07-02T08:57:16.193+0000    writing `testdb.testcollection` to `/backup/testdb/testcollection.bson`
2026-07-02T08:57:16.193+0000    writing `testdb.sample` to `/backup/testdb/sample.bson`
2026-07-02T08:57:16.194+0000    done dumping `testdb.testcollection` (2 documents)
2026-07-02T08:57:16.194+0000    done dumping `company.employee` (4 documents)
2026-07-02T08:57:16.194+0000    done dumping `testdb.sample` (1 document)
2026-07-02T08:57:16.199+0000    done dumping `test.employee` (5000 documents)
```

Restore toàn bộ database

```
test1@test1:~$ sudo mongorestore --drop /backup
2026-07-02T08:57:51.105+0000    preparing collections to restore from
2026-07-02T08:57:51.105+0000    don't know what to do with file `/backup/prelude.json`, skipping...
2026-07-02T08:57:51.105+0000    reading metadata for `company.employee` from `/backup/company/employee.metadata.json`
2026-07-02T08:57:51.105+0000    reading metadata for `test.employee` from `/backup/test/employee.metadata.json`
2026-07-02T08:57:51.106+0000    reading metadata for `testdb.sample` from `/backup/testdb/sample.metadata.json`
2026-07-02T08:57:51.106+0000    reading metadata for `testdb.testcollection` from `/backup/testdb/testcollection.metadat
a.json`
2026-07-02T08:57:51.106+0000    dropping collection `test.employee` before restoring
2026-07-02T08:57:51.108+0000    dropping collection `company.employee` before restoring
2026-07-02T08:57:51.110+0000    dropping collection `testdb.testcollection` before restoring
2026-07-02T08:57:51.110+0000    dropping collection `testdb.sample` before restoring
2026-07-02T08:57:51.211+0000    restoring `test.employee` from `/backup/test/employee.bson`
2026-07-02T08:57:51.239+0000    restoring `company.employee` from `/backup/company/employee.bson`
2026-07-02T08:57:51.261+0000    restoring `testdb.testcollection` from `/backup/testdb/testcollection.bson`
2026-07-02T08:57:51.282+0000    restoring `testdb.sample` from `/backup/testdb/sample.bson`
2026-07-02T08:57:51.292+0000    finished restoring `testdb.sample` (1 document, 0 failures)
2026-07-02T08:57:51.292+0000    finished restoring `company.employee` (4 documents, 0 failures)
2026-07-02T08:57:51.292+0000    finished restoring `testdb.testcollection` (2 documents, 0 failures)
2026-07-02T08:57:51.382+0000    finished restoring `test.employee` (5000 documents, 0 failures)
2026-07-02T08:57:51.382+0000    no indexes to restore for collection `company.employee`
2026-07-02T08:57:51.382+0000    no indexes to restore for collection `test.employee`
2026-07-02T08:57:51.382+0000    no indexes to restore for collection `testdb.testcollection`
2026-07-02T08:57:51.382+0000    no indexes to restore for collection `testdb.sample`
2026-07-02T08:57:51.382+0000    5007 document(s) restored successfully. 0 document(s) failed to restore.
```





Bật authentication + KeyFile

Tạo key file

máy test1

```
test1@test1:~$ openssl rand -base64 756 > mongodb-keyfile
test1@test1:~$ cat mongodb-keyfile
```

```
HLzQSSY0/8TOqHUv/a/qZrpdw9Uhd2vmaOEVsN5zsVPqELkkmkSxbiZoNlu4uA/A
IgqarEF6Re9orBH1W8ZvAAO6Mgmd4XSyFuyyYxKAu0eHZ+Z3im31NIbMtzAOKccP
dkck55x2h7+5GIklf9HrMaqO4kbMF5eLhsnRU8Ed2FaIt0y3tlEt+qcTtrq4IBPu
wxj80WKISgBFJ+UudteS5jaNgFIy2d2agpKCyjuRGRjE/hfpFK47ylWXxp+9FKbw
+DGHor2a+NOlxsAyom0+yIFR4b0PeBzDHh9qVSr+ih4BUy6/9qqv597EmU60CjgG
uItWf4FmjJvWapyIcZMmVg+bzoPkfTD7Y4OwSGuS/AqDtAlefaD+RJZxSLaEVPvA
LJbhp71ISPQt14Z7EiZIyOSQsxGIpAY6mfKb9G8X1bth5Sn1yvagjAb2Nt0boVJg
QyE6bkS1cqW+AT0H0MkO4LhyYQbk8E7wmMI4Ee1rL7vNVia8z3vkOfwct57nLQtM
fn0MJHw5bCn5Rs+b3TkPOPKXKEBsdW6ctl+gdDERLkrvJ0FyuvPH47afwTFrTJvf
yL1CAlOGrElbc2U/34S3GTsOyHxZROO4U4aPTD1Dt9GwvuIWS38eDobzpL2AFw2H
+YfwVYB+A/YHyzrwUD2WYa3XdZ2pQD+WNrc6wS4K21rcyfxLUMtS75taCXeroBjU
tODC/aqH+S37MPC9WC8WOzMk8ls8D3v9FS6Fu7y9s9mMN36jFisMb/ZWXSyVGCaX
RAN+BQN0pM3AMJK4jF/09ATsvInstGjnJcD8W66jiGd9kVVqG+viO/5/DsaJe84E
IzRUzbr/CdNoms1gZ3xJY2sj/J4yKfWvkjAfU1nvHeaKxa/xa0R8FWvy9N28HVg0
DcRswcYmD3DnnLEai87F8bKLJ+7yxfhIfMiJkyD1nCZwJdGMfZdHARb+xQYm/W5X
Mo5aJPa0hEjVDpMTeRDOaNn1sTiq0H8ftperD7729xjP/nsM
```


Copy keyfile từ test1 sang 2 máy test2,3

```
test1@test1:~$ scp mongodb-keyfile test2@172.31.1.242:/tmp/
The authenticity of host '172.31.1.242 (172.31.1.242)' can't be established.
ED25519 key fingerprint is: SHA256:iJvb/mVcoMwGl9e4636/qf93lgiOnIS6n4ZLSXWpYJQ
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.1.242' (ED25519) to the list of known hosts.
test2@172.31.1.242's password:
mongodb-keyfile                                              100% 1024     2.4MB/s   00:00
test1@test1:~$ scp mongodb-keyfile test3@172.31.1.246:/tmp/
The authenticity of host '172.31.1.246 (172.31.1.246)' can't be established.
ED25519 key fingerprint is: SHA256:VptbPDVhPvTvhIRALUr2/6/f9mdmfj07s7ULuteGgMY
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.1.246' (ED25519) to the list of known hosts.
test3@172.31.1.246's password:
mongodb-keyfile                                              100% 1024     2.4MB/s   00:00
```

Đưa keyfile vào đúng vị trí:

Test1 copy file vào vị trí + phân quyền

```
test1@test1:~$ sudo cp ~/mongodb-keyfile /etc/mongodb/
test1@test1:~$ sudo chown mongodb:mongodb /etc/mongodb/mongodb-keyfile
test1@test1:~$ sudo chmod 400 /etc/mongodb/mongodb-keyfile
test1@test1:~$ ls -l /etc/mongodb/mongodb-keyfile
-r-------- 1 mongodb mongodb 1024 Jul  3 01:36 /etc/mongodb/mongodb-keyfile
```

Test2, 3 chuyển file qua đúng vị trí + phân quyền

```
test2@test2:~$ sudo mkdir -p /etc/mongodb
[sudo: authenticate] Password:
test2@test2:~$ sudo mv /tmp/mongodb-keyfile /etc/mongodb/
test2@test2:~$ sudo chown mongodb:mongodb /etc/mongodb/mongodb-keyfile
test2@test2:~$ sudo chmod 400 /etc/mongodb/mongodb-keyfile
test2@test2:~$ ls -l /etc/mongodb/mongodb-keyfile
-r-------- 1 mongodb mongodb 1024 Jul  3 01:32 /etc/mongodb/mongodb-keyfile
```

```
test3@test3:~$ sudo mkdir -p /etc/mongodb
[sudo: authenticate] Password:
test3@test3:~$ sudo mv /tmp/mongodb-keyfile /etc/mongodb/
test3@test3:~$ sudo chown mongodb:mongodb /etc/mongodb/mongodb-keyfile
test3@test3:~$ sudo chmod 400 /etc/mongodb/mongodb-keyfile
test3@test3:~$ ls -l /etc/mongodb/mongodb-keyfile
-r-------- 1 mongodb mongodb 1024 Jul  3 01:32 /etc/mongodb/mongodb-keyfile
```

Chỉnh file mongo.conf

```
test1@test1:~$ sudo vim /etc/mongod.conf
test1@test1:~$ grep -n "security" -A5 /etc/mongod.conf
28:security:
29-  authorization: enabled
30-  keyFile: /etc/mongodb/mongodb-keyfile
31-
32-#operationProfiling:
```

```
test3@test3:~$ grep -n "security" -A5 /etc/mongod.conf
28:security:
29-  authorization: enabled
30-  keyFile: /etc/mongodb/mongodb-keyfile
31-
```

Restart system 2 máy test 2,3

```
test2@test2:~$ sudo systemctl restart mongod
test2@test2:~$ sudo systemctl status mongod
● mongod.service - MongoDB Database Server
     Loaded: loaded (/usr/lib/systemd/system/mongod.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-07-03 01:51:18 UTC; 5s ago
 Invocation: c5c8f654546b4edfa3d24a1713ef622d
       Docs: https://docs.mongodb.org/manual
   Main PID: 2378 (mongod)
     Memory: 183.3M (peak: 229.3M)
        CPU: 497ms
     CGroup: /system.slice/mongod.service
             └─2378 /usr/bin/mongod --config /etc/mongod.conf
```

Vào test1 test thử kết nối

```
rs0 [direct: other] test> rs.status().members.forEach(m => print(m.name, m.stateStr))
test1:27017 RECOVERING
test2:27017 (not reachable/healthy)
test3:27017 (not reachable/healthy)
```

Restart test 1 xong connect / Do authorized đã enable nên user test1 không thao tác được

```
rs0 [direct: primary] test>  rs.status().members.forEach(m => print(m.name, m.stateStr))
MongoServerError[Unauthorized]: Command replSetGetStatus requires authentication
```

Tạo user admin

```
rs0 [direct: primary] test> use admin
switched to db admin
rs0 [direct: primary] admin> db.createUser({
|   user: "admin",
|   pwd: "Admin@123",
|   roles: [
|     { role: "root", db: "admin" }
|   ]
| })
{
  ok: 1,
  '$clusterTime': {
```

```
rs0 [direct: primary] test> db.runCommand({ connectionStatus: 1 })
{
  authInfo: {
    authenticatedUsers: [ { user: 'admin', db: 'admin' } ],
    authenticatedUserRoles: [ { role: 'root', db: 'admin' } ]
  },
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1783044114, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('hFTE5fWy1xPSnFSbxFo7Bbl1TOk=', 0),
      keyId: Long('7657467780102356997')
    }
  },
  operationTime: Timestamp({ t: 1783044114, i: 1 })
```

Đăng nhập lại bằng user admin

```
test1@test1:~$ mongosh   -u admin   -p 'Admin@123'   --authenticationDatabase admin
Current Mongosh Log ID: 6a47230beff2eb569abccd8c
Connecting to:          mongodb://<credentials>@127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2
hSource=admin&appName=mongosh+2.9.2
Using MongoDB:          7.0.37
Using Mongosh:          2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-07-03T01:59:50.494+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage eng
ee http://dochub.mongodb.org/core/prodnotes-filesystem
------

rs0 [direct: primary] test>
```

Check lại lệnh

```
rs0 [direct: primary] test>  rs.status().members.forEach(m => print(m.name, m.stateStr))
test1:27017 PRIMARY
test2:27017 SECONDARY
test3:27017 SECONDARY
```

Tạo user chỉ được đọc/ghi database company:

```
rs0 [direct: primary] test> use company
switched to db company
rs0 [direct: primary] company> db.createUser({
|   user: "appuser",
|   pwd: "123",
|   roles: [
|     { role: "readWrite", db: "company" }
|   ]
| })
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1783044151, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('CY20HJgIIUEs0AgQxAbZG3vJZ0A=', 0),
      keyId: Long('7657467780102356997')
    }
  },
  operationTime: Timestamp({ t: 1783044151, i: 1 })
```

Đăng nhập bằng user mới tạo

```
test1@test1:~$ mongosh \
  -u appuser \
  -p '123' \
  --authenticationDatabase company
Current Mongosh Log ID: 6a4724c283db108eead909d8
Connecting to:          mongodb://<credentials>@127.0.0.1:27017/?directConnection=tru
hSource=company&appName=mongosh+2.9.2
Using MongoDB:          7.0.37
Using Mongosh:          2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

rs0 [direct: primary] test>
```

Kiểm tra database

Trên admin thấy toàn bộ db

```
rs0 [direct: primary] test> show dbs
admin     220.00 KiB
company    40.00 KiB
config    256.00 KiB
local       4.90 MiB
test      308.00 KiB
testdb     80.00 KiB
```

Trên user appuser chỉ thấy đc db cho phép

```
rs0 [direct: primary] test> show dbs
company  40.00 KiB
```

Kiểm tra đọc dữ liệu/ appuser

```
rs0 [direct: primary] test> use company
switched to db company
rs0 [direct: primary] company> db.employee.find().limit(5)
[
  {
    _id: ObjectId('6a461ecfb4fdb15d62def134'),
    name: 'Nguyen Van A',
    position: 'DevOps Engineer',
    department: 'IT',
    salary: 1500
  },
  {
    _id: ObjectId('6a461ecfb4fdb15d62def135'),
```

Kiểm tra ghi dữ liệu/ appuser

```
rs0 [direct: primary] company> db.employee.insertOne({
|     name: "Hung",
|     dept: "IT",
|     createdAt: new Date()
| })
{
  acknowledged: true,
  insertedId: ObjectId('6a47255683db108eead909d9')
}
rs0 [direct: primary] company>
```

```
  {
    _id: ObjectId('6a47255683db108eead909d9'),
    name: 'Hung',
    dept: 'IT',
    createdAt: ISODate('2026-07-03T02:58:30.580Z')
  }
]
```

Ghi dữ liệu thành công!

User khác bị chặn truy cập / read hay insert

```
rs0 [direct: primary] company> use test
switched to db test
rs0 [direct: primary] test> db.employee.find()
MongoServerError[Unauthorized]: not authorized on test to execute comman
ID("77bcdc4c-ba69-48eb-91e9-554458f706a0") }, $clusterTime: { clusterTi
BinData(0, D481A13E1506B13D9A237A4224F30B390D506B17), keyId: 76574677801
referred" }, $db: "test" }
rs0 [direct: primary] test> db.employee.insertOne({a:1})
MongoServerError[Unauthorized]: not authorized on test to execute comman
: ObjectId('6a4725eb83db108eead909da') } ], ordered: true, lsid: { id: U
xnNumber: 1, $clusterTime: { clusterTime: Timestamp(1783047601, 1), sign
4224F30B390D506B17), keyId: 7657467780102356997 } }, $readPreference: {
rs0 [direct: primary] test>
```

Kiểm tra quyền user:

```
rs0 [direct: primary] test> use company
switched to db company
rs0 [direct: primary] company> db.getUser("appuser")
{
  _id: 'company.appuser',
  userId: UUID('f2ed430e-d11c-4402-aa2c-a645a3dc98a9'),
  user: 'appuser',
  db: 'company',
  roles: [ { role: 'readWrite', db: 'company' } ],
  mechanisms: [ 'SCRAM-SHA-1', 'SCRAM-SHA-256' ]
}
```

Mornitoring

theo dõi thông tin node

```
rs0 [direct: primary] test> rs.status().members.forEach(m =>
|     print(
|         "Node:", m.name,
|         "| State:", m.stateStr,
|         "| Health:", m.health,
|         "| Uptime:", m.uptime + "s"
|     )
| )
Node: test1:27017 | State: PRIMARY | Health: 1 | Uptime: 4786s
Node: test2:27017 | State: SECONDARY | Health: 1 | Uptime: 2361s
Node: test3:27017 | State: SECONDARY | Health: 1 | Uptime: 2361s
```

Theo dõi Replication:

```
rs0 [direct: primary] test> rs.printReplicationInfo()
actual oplog size
'2630.0693359375 MB'
---
configured oplog size
'2630.0693359375 MB'
---
log length start to end
'155710 secs (43.25 hrs)'
---
oplog first event time
'Wed Jul 01 2026 08:05:51 GMT+0000 (Coordinated Universal Time)'
---
oplog last event time
'Fri Jul 03 2026 03:21:01 GMT+0000 (Coordinated Universal Time)'
---
now
'Fri Jul 03 2026 03:21:01 GMT+0000 (Coordinated Universal Time)'
```

Oplog càng nhỏ khả năng lỗi càng cao

```
rs0 [direct: primary] test> rs.printSecondaryReplicationInfo()
source: test2:27017
{
  syncedTo: 'Fri Jul 03 2026 03:22:11 GMT+0000 (Coordinated Universal Time)',
  replLag: '0 secs (0 hrs) behind the primary '
}
---
source: test3:27017
{
  syncedTo: 'Fri Jul 03 2026 03:22:11 GMT+0000 (Coordinated Universal Time)',
  replLag: '0 secs (0 hrs) behind the primary '
}
```

replLag cao tầm >10s đang bị chậm

Connection

```
rs0 [direct: primary] test> db.serverStatus().connections
{
  current: 36,
  available: 20303,
  totalCreated: 127,
  rejected: 0,
  active: 7,
  queuedForEstablishment: Long('0'),
  establishmentRateLimit: {
    rejected: Long('0'),
    exempted: Long('0'),
    interruptedDueToClientDisconnect: Long('0')
  },
  threaded: 36,
  exhaustIsMaster: 0,
  exhaustHello: 5,
  awaitingTopologyChanges: 5
}
rs0 [direct: primary] test>
```

Current: client đang kết nối

Available: còn bao nhiêu connection

Opcounters

```
rs0 [direct: primary] test> db.serverStatus().opcounters
{
  insert: Long('3'),
  query: Long('77'),
  update: Long('39'),
  delete: Long('0'),
  getmore: Long('1372'),
  command: Long('17999')
}
rs0 [direct: primary] test>
```

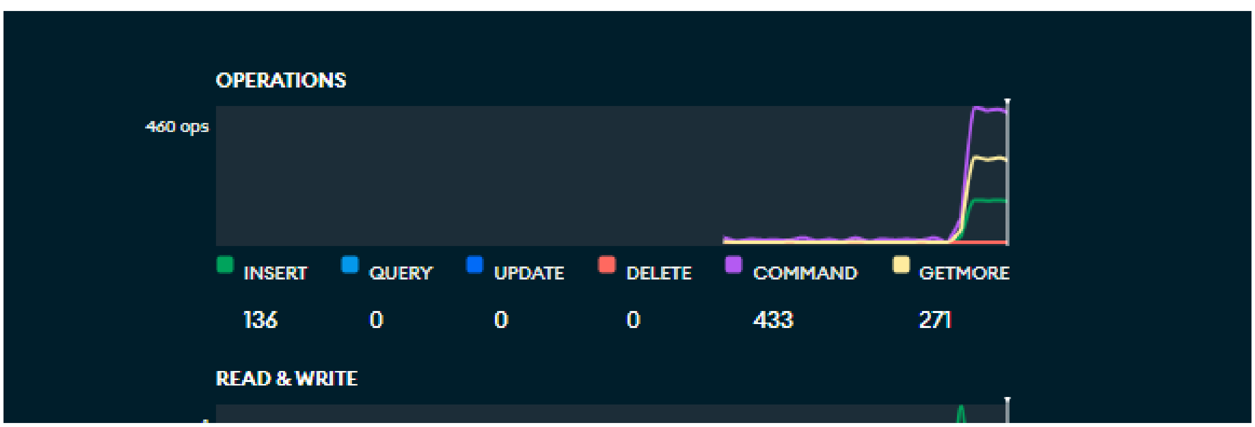
Hệ thống đang thực hiện thao tác nào nhiều!!

Test insert data hàng loạt

```
rs0 [direct: primary] company> for(let i=0;i<5000;i++){
|
| db.employee.insertOne({
|
| name:"Hung",
|
| age:i
|
| })
|
| }
{
  acknowledged: true,
  insertedId: ObjectId('6a472df07b7cf2e9a4aa442a')
}
rs0 [direct: primary] company>
```

```
rs0 [direct: primary] test> db.serverStatus().opcounters
{
  insert: Long('540'),
  query: Long('81'),
  update: Long('40'),
  delete: Long('0'),
  getmore: Long('2541'),
  command: Long('21249')
}
rs0 [direct: primary] test>
```

Xem trên MongoDB compass




OPERATIONS
469 ops
INSERT 0
QUERY 0
UPDATE 0
DELETE 0
COMMAND 6
GETMORE 0
READ & WRITE
1
AREADS 0
AWRITES 0
QREADS 0
QWRITES 0
NETWORK
607 KB
41 conn
BYTESIN 2
BYTESOUT 97
CONNECTIONS 41